Quiz 1

Graded

Student

Colin Skinner

Total Points

18 / 20 pts

Question 1

Instructions

0 / 0 pts

✔ **+ 0 pts** Correct

+ 0 pts Incorrect

Question 2

(no title)

5 / 5 pts

✔ **+ 5 pts** **Correct**: 0.00028

+ 5 pts **Rounding**: We awarded full points since your answer is correct, but note for next time to round after you finish computing all the values (such that the answer is 0.00028).

+ 5 pts **Showing work**: We awarded full points since your answer is correct, but for next time, make sure to include more of your work as we may not give full points for only an answer shown.

+ 4.5 pts **Small computational mistake**: Work is mostly correct, but there is a small computational error.

+ 4 pts **Two small computational mistakes**: Work is mostly correct and clearly shown, but there are two small computational errors.

+ 3.5 pts **Incorrect or lack of weighting of** $P(X_1)$: Should be $P(X_1) = \frac{1}{10}$.

+ 3.5 pts **Denominator calculated incorrectly**: See comments

+ 2.5 pts **Partial**: Some work shown; right direction.

+ 0 pts No answer

+ 0 pts ****other

– 0.5 pts Computational error

Question 3

(no title)

3 / 5 pts

+ 5 pts Correct

+ 4 pts **Incorrect final answer:** rounding or arithmetic error

+ 2.5 pts **Inconclusive answer:** multiple answers given or non-numerical answer

✔ **+ 3 pts** **Incorrect answer:** plugged in $P(a^1)$ instead of the value of a (which is 1).

+ 2 pts **Partial Credit**: Included $0.5(1) + 0.2$ or $0.5(a) + 0.2$ somewhere in work shown

+ 1 pt **Partial Credit**: Made attempt of calculating the weight of the sample (but went down incorrect route)

+ 0 pts Incorrect

Close! We do not want to use P(a=1) though, we want to use the value of a, which is 1

Question 4

(no title) **5** / 5 pts

✔ **+ 5 pts** Correct

+ 0 pts No answer

+ 4 pts Typo or transcription error

+ 3 pts Incorrect mean or mode

+ 2 pts Mean and mode are vectors, not a single value

+ 2 pts Correct posterior distribution, but incorrect mean and mode

+ 4 pts Mean and mode vectors are not Dir distribution

+ 2 pts Incorrect posterior distribution, but followed the right procedure to calculate mean and mode

+ 0 pts Incorrect

Question 5

(no title) **5** / 5 pts

✔ **+ 5 pts** Correct

+ 0 pts Incorrect

+ 2.5 pts ambiguous reasoning

Q1 Instructions
0 Points

You may consult any material (e.g., books, calculators, computer programs, and online resources), but you may not consult other people inside or outside of the class or large language models like *ChatGPT*.

The quiz is designed to be completed in 60 minutes, but we will grant you 90 minutes total to complete and submit your quiz (including uploading any images, handling any logistical issues, etc.) The timing on Gradescope is a hard cutoff.

Ed will not allow any public posts during that time. Partial credit may be awarded based on work shown. You can upload **either** a file **or** give a response in the free response field (LaTeX is possible using *two* $ signs before and after your math expression – so a total of four $ signs).

Out of fairness to all students, only material submitted during the allowed time will be graded. Also, if you upload a file (only pdf or image files, please), please *do not* include identifying information as the exam is graded anonymously.

As a reminder, all questions are equally weighted. Some questions will take longer than others. Feel free to answer the questions in any order you like.

Please sign here with your name, acknowledging that you are aware of the honor code and that you comply with the honor code.

Write your name in the box below.

Colin Skinner

Q2
5 Points

We represent a joint distribution over a sequence of four digits $(0–9)$ as a Bayesian network. The ith digit is denoted X_i. The structure of the network is such that there are only directed edges from X_i to X_{i+1} for $i = 1, 2, 3$ with

$$P(x_{i+1}^j \mid x_i^k) = \frac{(j-k)^2}{\sum_{\ell=0}^{9}(\ell-k)^2}$$

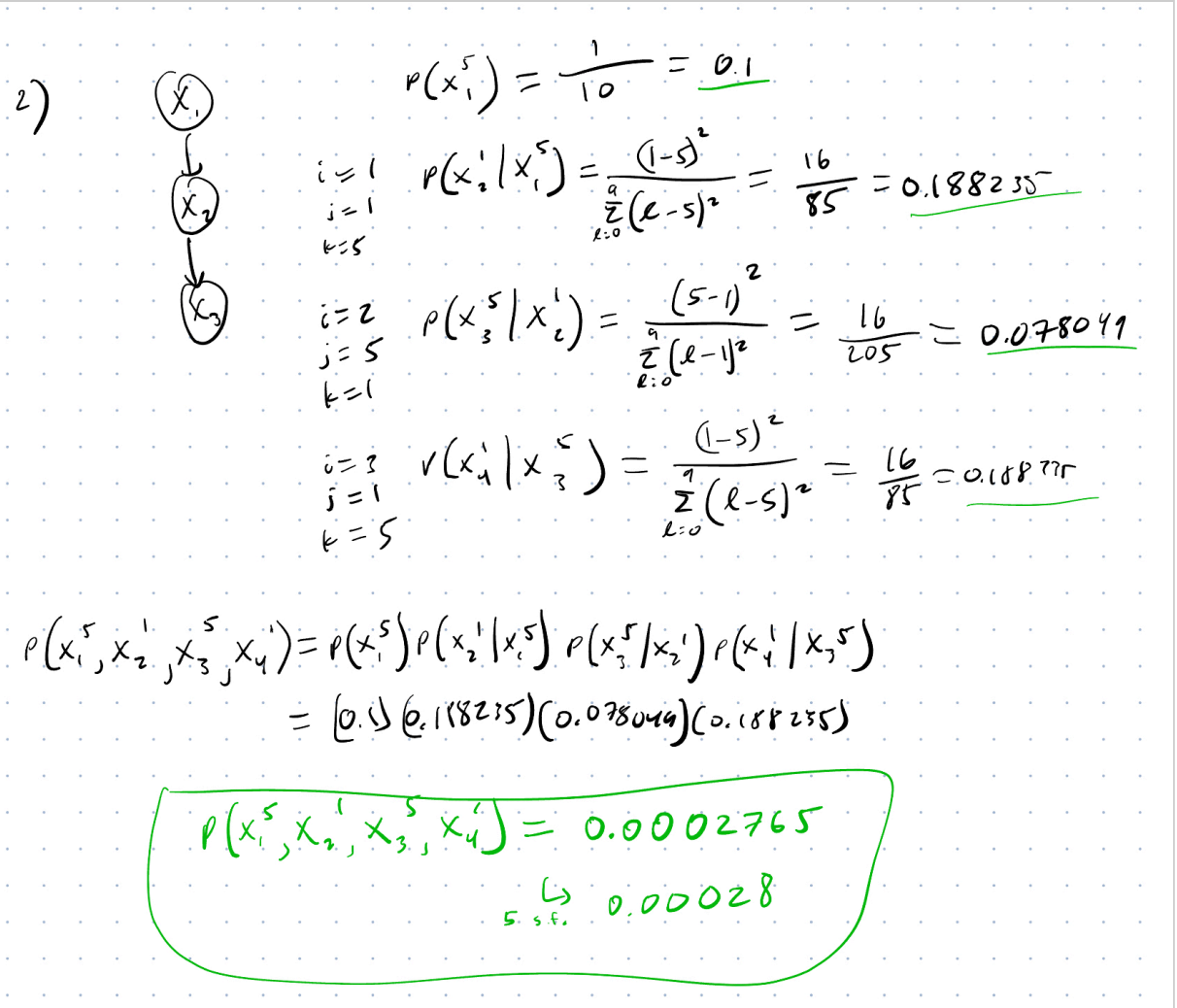
The distribution over X_1 is uniform. What is the probability of the sequence **5151** (meaning $X_1 = 5$, $X_2 = 1$, $X_3 = 5$, $X_4 = 1$)?

To help clarify notation regarding the expression $P(x_{i+1}^j \mid x_i^k)$ above: if we have $P(x_2^1 \mid x_1^5)$, that means $X_1 = 5$ and $X_2 = 1$.

Provide your answer rounded to five decimal places, such as 0.00001 or 0.02535. If the actual value is 0.025346, you should enter 0.02535.

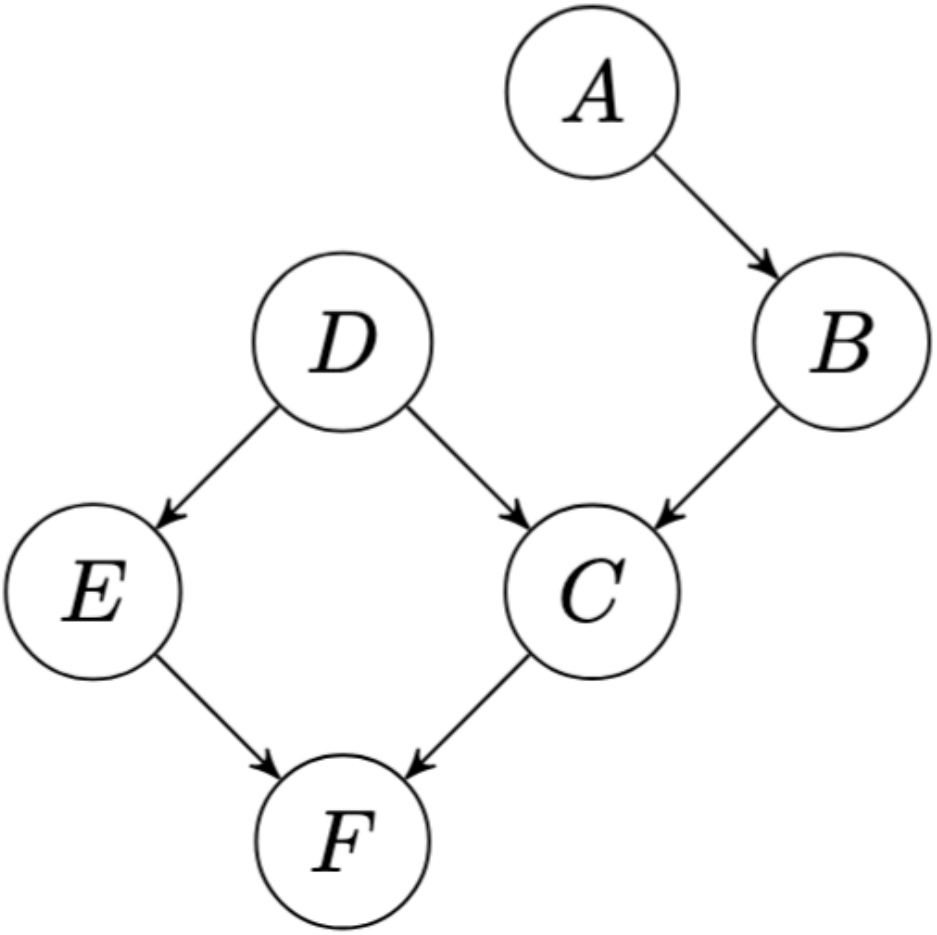
0.00028

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Q3
5 Points

Suppose we have a Bayesian network consisting of binary variables with the following structure:



We have the following:

	Probability
$P(a^1)$	0.9
$P(b^1 \mid a)$	$0.5a + 0.2$
$P(d^1)$	0.2
$P(c^1 \mid b, d)$	$0.1b + 0.2d + 0.1$
$P(e^1 \mid d)$	$0.2d + 0.2$
$P(f^1 \mid c, e)$	$0.2c + 0.2e + 0.3$

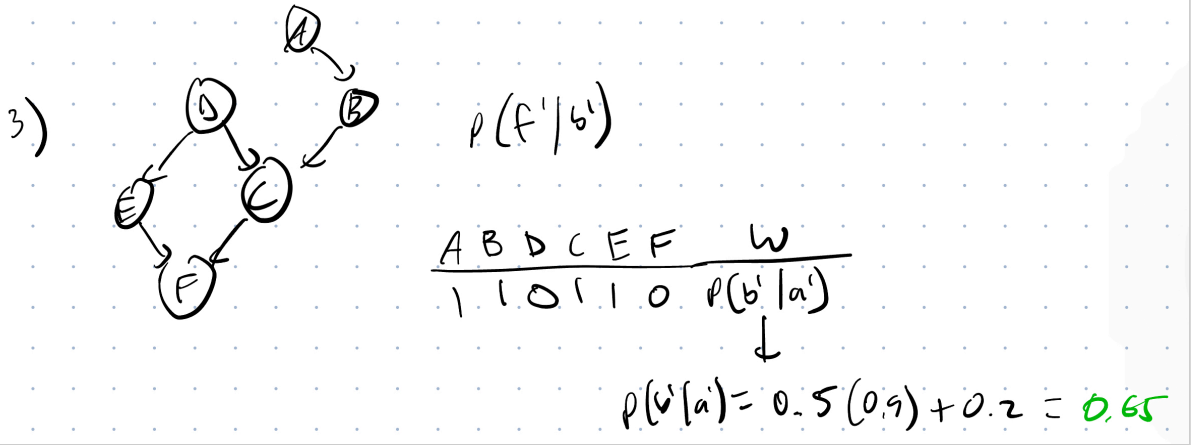
We want to apply likelihood weighted sampling to estimate $P(f^1 \mid b^1)$.

What is the **weight** of the sample $(a^1, b^1, c^1, d^0, e^1, f^0)$?

Note that we are **not** looking for the joint distribution of this sample, but the **weight of the sample**.

0.65

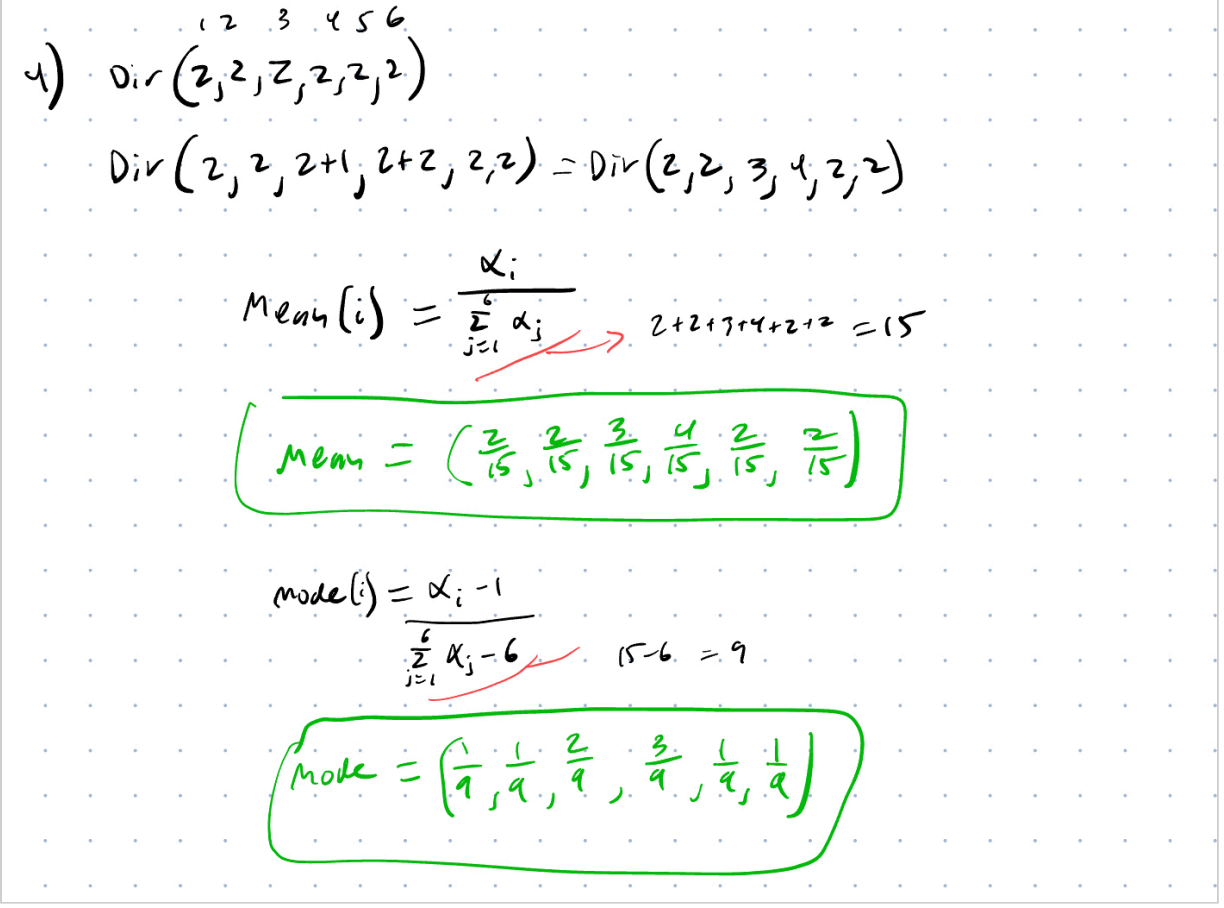
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Q4
5 Points

Johnny hands you a strange six-sided die with the sides labeled 1 through 6. The construction of it indicates to you that it is not fair, but you start with a Dirichlet prior $\mathrm{Dir}(2, 2, 2, 2, 2, 2)$ to capture your initial uncertainty about the outcome distribution. You roll it three times and get a 4, 3, and 4. Write down the mean and the mode of your posterior distribution.

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Q5
5 Points

How does $K2$ search avoid creating cycles in a Bayesian network? Answer in 1-3 sentences.

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